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CARES Technical Approval Report TA1C-5003

Assessment of the Erico Lenton® Taper Threaded Coupler Product, BNFL type B Couplers and the Quality System for Production

LENTON®

Mechanical Rebar Splicing System



ERICO®

CARES TECHNICAL APPROVAL

UKAS PRODUCT CERTIFICATION 5003

Product

ERICO LENTON® A12, P13 and P14 mechanical couplers for reinforcing steel

Produced by:

ERICO BV,
Jules Verneweg 75,
5015 BG,
Tilburg,
The Netherlands

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1 Product Summary

The products are for the mechanical connection of deformed high-yield carbon steel bars for the reinforcement of concrete complying with the requirements of BS4449 grade B500B and grade B500C.

1.1 Scope of Application

ERICO LENTON® A12, P13 and P14 taper threaded couplers, in the size range 12mm-40mm, have been evaluated for use as follows:

- Static tension or compression applications in reinforced concrete structures designed to BS8110: Part 1 in accordance with CARES Appendix TA1-B using grade B500B and B500C reinforcement.
- Static tension applications in accordance with Type B BNFL Specification for Couplers and CARES Appendix TA1-C using B500C reinforcement only.

1.2 Design Considerations

BS 8110 Clause 3.12.8.9 gives guidance on the positioning of couplers.

The specified cover for fire resistance and durability should be provided to the coupler sleeve. All couplers have been designed with controlled mechanical properties to be compatible with reinforcing bars to BS4449.

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1.3 Conclusion

It is the opinion of UK CARES that ERICO LENTON® A12, P13 and P14 taper threaded couplers are satisfactory for use within the stated limits when applied and used in accordance with the manufacturer's instructions and the requirements of this certificate.



B. Bowsher
Executive Director

July 2007

2 Technical Specification

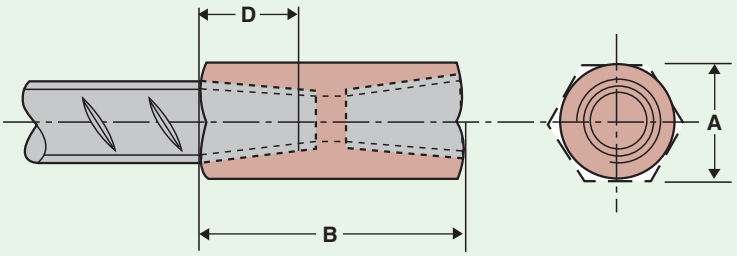
2.1 General

ERICO LENTON® A12, P13 and P14 taper threaded couplers are for joining deformed Grade B500B and Grade B500C steel reinforcing bars complying with BS 4449.

2.2 LENTON® Type A12

The standard A12 couplers are designed for use where one of the bars to be coupled can be rotated. The 12 mm to 20 mm couplers are a one-piece hexagonal section coupler with a tapered thread machined in each end. The 25 mm to 40 mm diameter couplers are a one-piece circular section with a tapered thread machined in each end.

Standard A12 Coupler



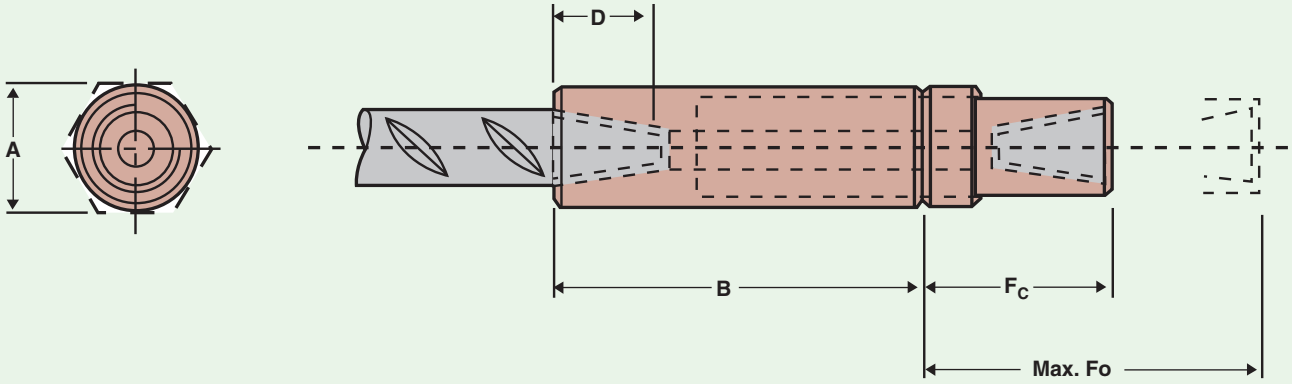
2.3 Types P13 and P14

The P13 and P14 three-piece positional couplers are for use where neither of the bars to be coupled can be rotated. The P13 coupler is specifically used when the bar to be coupled are also restricted in their axial movement. The three components comprise male and female ends with internal taper threads, connected by a female and male parallel threads and locknut.

The male section is hexagonal in cross section for 12 mm to 20 mm couplers. The female section is circular in cross section for 25 to 40 mm. The female section is hexagonal in cross section for 12 to 16 mm couplers and circular in cross section for 20 to 40 mm.

KEY
A = diameter
B = length of coupler body
D = bar engagement
Fc = connector and jam nut (closed position) length
Max. Fo = connector and jam nut (fully open position installed) length

Positional P13 Coupler



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The dimensions are as follows:

Size mm	Part No	Art No	"A" mm	"B" mm	"D" mm	Weight kg
12	EL12A12	150010	17*	49	19	0.06
16	EL16A12	150030	22*	61	24	0.12
20	EL20A12	150050	27*	88	36	0.26
25	EL25A12	150070	33	96	41	0.35
32	EL32A12	150100	42	107	46	0.60
40	EL40A12	150140	52	131	58	1.12

**Use hexagonal material (measured across the flats), others use round material*

The dimensions for the P13 coupler are as follows:

Size mm	Part No	Art No	"A" mm	"B" mm	"Fc" mm	"Fo" mm	"D" mm	Weight kg
12	EL12P13	150190	22*	77	42	85	19	0.29
16	EL16P13	150210	27*	88	48	93	24	0.50
20	EL20P13	150230	33	112	63	116	36	0.87
25	EL25P13	150250	41	127	68	128	41	1.31
32	EL32P13	150280	52	144	79	144	46	2.38
40	EL40P13	150320	64	179	91	172	58	4.34

**Use hexagonal material (measured across the flats), others use round material*

The dimensions for the P14 coupler are as follows:

Size mm	Part No	Art No	"A" mm	"B" mm	"Fc" mm	"Fo" mm	"D" mm	Weight kg
12	EL12P14	150370	22*	45	42	53	19	0.20
16	EL16P14	150390	27*	54	48	59	24	0.35
20	EL20P14	150410	33	75	63	79	36	0.64
25	EL25P14	150430	41	83	68	84	41	0.96
32	EL32P14	150460	52	95	79	95	46	1.77
40	EL40P14	150500	64	114	91	107	58	3.07

**Use hexagonal material (measured across the flats), others use round material*

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3 Product Performance and Characteristics

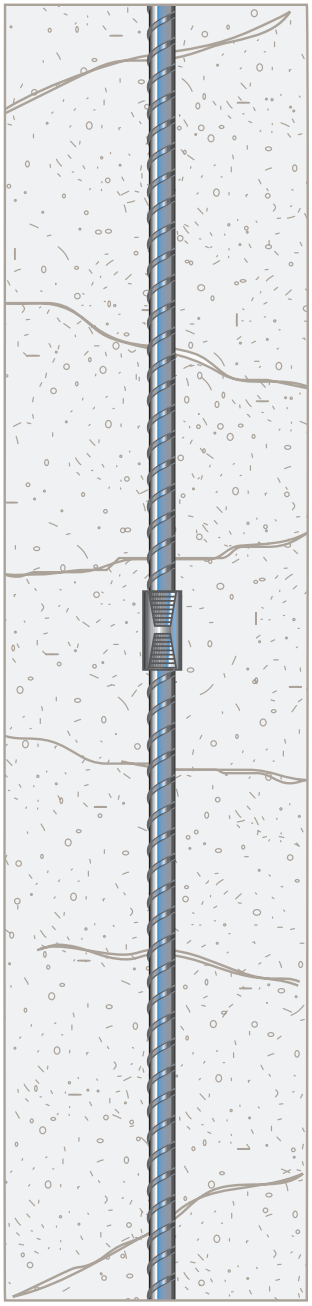
Full destructive tests have been carried out to demonstrate compliance with performance requirements defined in CARES Appendix TA1-B and Appendix TA1-C including the performance requirements of BS8110 and the BNFL Specification for Couplers when used with reinforcing bars to BS4449:

CARES APPENDIX TA1-B

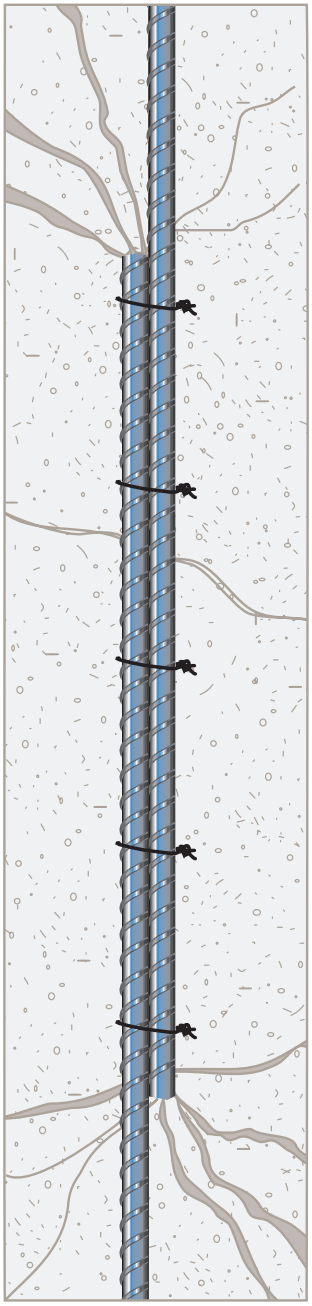
- Permanent deformation is less than 0.1mm at 300 MPa in tension.
- Permanent deformation is less than 0.1mm at 300 MPa in compression.
- Tensile strength is greater than 540 MPa for Grade B500B and 575 MPa for Grade B500C.
- Compressive strength is greater than 540 MPa for Grade B500B and 575 MPa for Grade B500C.

CARES APPENDIX TA1-C

- Permanent deformation is less than 0.1mm at 300 MPa in tension.
- Tensile strength is greater than 575 MPa for Grade B500B and 575 MPa for Grade B500C and greater than the load required to produce 2% strain in the reference bars.
- Cyclic loading of 100 cycles between 5% and 90% f_y.
- Reduced temperature performance at -7ºC.



Mechanical splicing provides the assurance of maintaining load path continuity of the structural reinforcement independent of the condition or existence of the concrete



Lap splices rely on bond with the concrete for effective continuity of reinforcement, which can result in localised areas of increased concrete stress that must be considered by the designer

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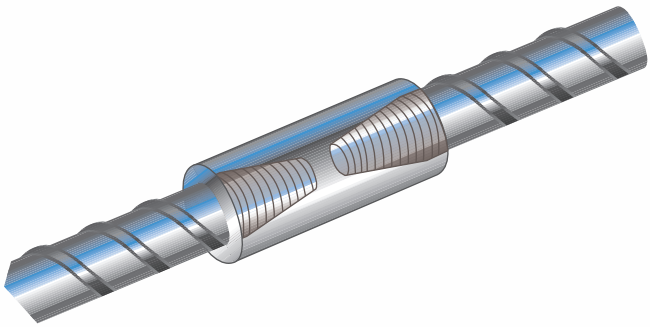
4 Installation

The bars to be threaded must be cut square and threaded, using ERICO LENTON® equipment and suitably trained and experienced operatives in accordance with LENTON® operating instructions. These operators will have an ERICO LENTON® Training Certificate.

It is essential that the joints be tightened to the correct torque mentioned in the installation manual, using the appropriate LENTON® torque wrench.

4.1 Type A12 Coupler

A12 Couplers are for connecting reinforcing bars where one of the bars forming the splice is free to turn. The parts are screwed together by hand and tightened to the correct torque. Where the coupler forms a stop end, it must be torque tightened to the reinforcing bar. When the joint is to be made the continuation bar is screwed into the coupler by hand and tightened to the correct torque. The continuation bar must be able to rotate freely.



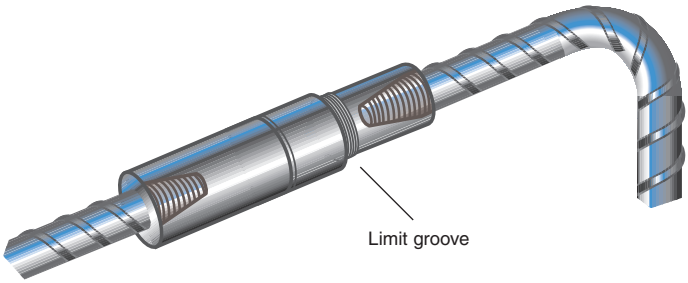
A12 Coupler

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4.2 Type P13 Coupler

P13 Couplers are for connecting cages, crowns of arches, or bars positioned and essentially unable to move. The P13 coupler can be adjusted in length to connect to the second bar. It must be ensured that the coupler is fully screwed together and tightened onto the first bar. The cage or other bar is brought into its final position ensuring that the gap between it and the coupler is not greater than 10-25 mm depending on the used size of rebar. The male part is screwed out to engage the tapered thread of the ongoing bar and then fully tightened. The lock-nut is spun back and fully tightened against the body of coupler. It must be ensured that the limit groove on the male section is not fully visible above the lock-nut.



P13 & P14 Couplers

4.3 Type P14 Coupler

P14 Couplers are for connecting onto a bent bar or a bar that cannot be rotated. It must be ensured that the male part is fully screwed into the female part and the female part is tightened onto the first bar. The taper-threaded bent bar is inserted into the mouth of the coupler as far as possible. The male part is screwed out of the female part to engage the taper-threaded bar and is tightened. The lock-nut is spun back to coupler body and fully tightened against the coupler body after setting the bent bar to the required position. It must be ensured that the limit groove is not fully visible above the lock-nut.

5 Safety Considerations

Care must be taken in handling and installing couplers. Couplers are supplied in containers that have a maximum weight of 25 kg. It is advisable to wear protective gloves during handling the containers, threaded bars and installing the couplers.

6 Product Testing and Evaluation

ERICO LENTON® taper threaded couplers have been tested to satisfy the requirements of BS8110 and BNFL Specification for Couplers, when used with reinforcing bars to BS4449. The testing comprised the following elements:

BS8110

- Compressive Strength
- Tensile Strength
- Permanent Deformation in tension and compression

BNFL Specification for Couplers

- Tensile Strength*
- Permanent Deformation*
- Cyclic Loading

* Low temperature testing at -7°C was included

Testing has been undertaken at Bureau Veritas.

The products are subject to a programme of periodic testing to ensure that they remain within the performance limits of this technical approval.

7 Quality Assurance

ERICO LENTON® taper thread couplers are produced under an ISO9001 quality assurance system certified by an independent certification body. The manufacture of the couplers has a CARES ISO9002 certificate. The quality assurance scheme monitors the production of the couplers and ensures that materials and geometry remain within the limits of this technical approval.

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8 Building Regulations

8.1 The Building Regulations (England and Wales)

Structure, Approved Document A

ERICO LENTON® A12, P13 and P14 taper threaded couplers, when used in BS8110 based designs using the data contained within this technical approval, satisfy the relevant requirements of The Building Regulations (England and Wales), Approved Document A.

Materials and Workmanship, Approved Document, to support regulation 7

This technical approval gives assurance that the ERICO LENTON® taper threaded couplers comply with the material requirements of BS8110.

8.2 The Building Regulations (Northern Ireland)

Part B, Materials and Workmanship

This technical approval gives assurance that ERICO LENTON® taper threaded couplers comply with the material requirements of BS8110 by virtue of regulation *B3, Deemed to satisfy provisions regarding the fitness of materials and workmanship.*

8.3 The Building Standards (Scotland) Regulations

Part B, Fitness of Materials

This technical approval gives assurance that ERICO LENTON® taper threaded couplers comply with the material requirements of BS8110 by virtue of *Clause B2.1*

Part C, Structure

ERICO LENTON® taper threaded couplers, when used in BS8110 based designs using the data contained within this technical approval, satisfy the requirements of *The Building Standards (Scotland) Regulations 1990, Part C, C2.1 clause b. construction,ii.*

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9 References

- BS 4449: 2005: Steel for the reinforcement of concrete - Weldable reinforcing steel - Bar, coil and decoiled product - Specification
- BS 8110: Part 1: 1997 (Revised 2005): Structural Use of Concrete, Code of Practice for Design and Construction
- BS EN ISO 9001: 2000: Quality management systems - Requirements
- BNFL Technical Standard: A.0391_1: Mechanical Splices to Reinforcement for Concrete
- TA1-B Quality and Operations Schedule for the Technical Approval of Couplers for Reinforcing Steel for BS810 Applications for Static Tension or Static Compression
- TA1-C Quality and Operations Schedule for the Technical Approval of Tension Couplers for Reinforcing Steel for BNFL Standard Applications

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10 Conditions

1. The quality of the materials and method of manufacture have been examined by CARES and found to be satisfactory. This technical approval will remain valid provided that:

   a The product design and specification is unchanged.
   b The materials and method of manufacture are unchanged.
   c The manufacturer complies with CARES regulations for technical approvals.
   d The manufacturer holds a valid CARES Certificate of Product Assessment.
   e The product is installed and used as described in this report.

2. CARES make no representation as to the presence or absence of patent rights subsisting in the product and/or the legal right of ERICO BV to market the product.

3. Any references to standards, codes or legislation are those which are in force at the date of this certificate.

4. Any recommendations relating to the safe use of this product are the minimum standards required when the product is used. These requirements do not purport to satisfy the requirements of the Health and Safety at Work act 1974 or any other relevant safety legislation.

5. CARES does not accept any responsibility for any loss or injury arising as a direct or indirect result of the use of this product.

6. This Technical Approval Report should be read in conjunction with CARES Certificate of Product Assessment No 5003. Confirmation that this technical approval is current can be obtained from UK CARES.

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